

PERSONAL REFLECTION & FEEDBACK FORM

BIL 476 / BIL 573 — Data Mining Course Project

TOBB University of Economics and Technology — Spring 2026

This form is submitted as a SEPARATE PDF alongside the main report.

Full Name	Erdem Baran
Student ID	211101034
Project Title	Market Basket Analysis on Online Retail Data: A Comparative Study of Apriori and FP-Growth Algorithms with Multi-Measure Rule Evaluation
Chosen Topic	Topic 1 — Market Basket Analysis
Date	27/03/2026

1. WHAT DID YOU LEARN?

Top 3 data mining concepts learned or reinforced:

1. Association rule mining fundamentals — understanding the difference between support, confidence, and lift, and how each metric captures a different aspect of rule interestingness. Applying these measures to real transaction data made the theoretical definitions concrete.
2. Algorithm efficiency and scalability — through direct comparison of Apriori and FP-Growth across multiple support thresholds, I gained a clear understanding of how candidate generation becomes a computational bottleneck at low support values, and why tree-based approaches like FP-Growth overcome this limitation.
3. Null-invariant interestingness measures — I learned why confidence and lift can be misleading in sparse datasets, and how Kulczynski and cosine measures address the null-invariance problem by ignoring transactions where neither item appears.

Practical skills developed:

Python data preprocessing with pandas (handling missing values, filtering, type conversion), association rule mining with mlxtend (Apriori and FP-Growth implementations), data visualization with matplotlib and seaborn, IEEE-format academic report writing with LaTeX on Overleaf, and version control with Git/GitHub.

2. CHALLENGES FACED

Challenge 1: Large dataset size — the raw xlsx file (541,909 rows) took considerable time to load and process in memory.

How addressed: Focused analysis on UK transactions only (91.4% of data), which reduced the transaction matrix to a manageable size while preserving analytical validity. Resolved: Yes.

Challenge 2: High sparsity of the transaction matrix (99.5%) — at low minimum support thresholds, Apriori became extremely slow (141 seconds at 1% support).

How addressed: Used FP-Growth as the primary algorithm for rule mining and kept the main analysis at 2% support. Conducted scalability comparison to document and explain the difference. Resolved: Yes.

Challenge 3: Missing CustomerID (24.93%) — a large proportion of transactions had no customer identifier, requiring a decision on how to handle them.

How addressed: Removed rows with missing CustomerID since they could not be attributed to a basket owner. This choice was documented and justified in the report's preprocessing section. Resolved: Yes.

Most difficult concept: Understanding the null-invariance problem and why measures like lift can produce misleading results for asymmetric item frequencies.

3. TOOLS & AI USAGE

Programming language: Python 3.10

Key libraries: pandas, numpy, mlxtend, matplotlib, seaborn, scipy

Report writing tool: LaTeX (Overleaf)

AI tool usage:

☒ Yes

☐ No

Tool used: Claude (Anthropic)

Purpose:

☒ Understanding concepts

☐ Debugging

☒ Code generation

☒ Report writing

☒ LaTeX help

Approximate % of code that was AI-assisted:

☐ 0%

☐ 1-25%

☐ 26-50%

☒ 51-75%

☐ 76-100%

Approximate % of report text that was AI-assisted:

☐ 0%

☒ 1-25%

☐ 26-50%

☐ 51-75%

☐ 76-100%

Verification method: All AI-generated code was executed and outputs were manually reviewed for correctness. Statistical results (support, confidence, lift values) were cross-checked against the raw data. Report text was reviewed, revised and written primarily in my own words, with AI used only for structural suggestions and LaTeX formatting help.

4. REFLECTION

Most interesting or surprising finding:

The near-zero correlation between support and lift ($r = -0.014$) was the most surprising result. I expected high-support rules to also have high lift, but the data showed that the most interesting co-purchasing patterns (such as the Regency Teacup set with lift = 24.12) actually had relatively low support values. This highlights that filtering rules by support alone would eliminate the most meaningful associations.

If I could start over, I would:

Apply a lower minimum support threshold (0.5-1%) from the start using FP-Growth, to discover a richer set of rules including rarer but potentially more actionable patterns. I would also explore multi-level association rules using product category hierarchies to find patterns at different levels of abstraction.

5. COURSE FEEDBACK

Project difficulty:

- ☐ Too Easy
- ☒ Appropriate
- ☐ Too Hard

Project duration:

- ☐ Too Short
- ☒ Appropriate
- ☐ Too Long

Instructions clarity:

- ☒ Clear
- ☐ Somewhat Unclear
- ☐ Unclear

Most useful course topics for this project:

Association rule mining (Apriori, FP-Growth), interestingness measures (support, confidence, lift, Kulczynski, cosine), and the discussion of the null-invariance problem. The data preprocessing lectures were also directly applicable.

Suggestions for future students:

Starting with a smaller dataset or higher support threshold for initial exploration helps understand the algorithms before scaling up. The mlxtend library documentation is comprehensive and recommended as a first reference.

ACADEMIC INTEGRITY STATEMENT

I declare that:

1. This project is entirely my own individual work.
2. All sources and tools are properly cited in the report.
3. Results are genuine and have not been fabricated.
4. AI tool usage has been honestly reported above.

Signature: Erdem Baran

Date: 27/03/2026